Technology Stack

Date	23 June 2026
Team ID	XXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, JavaScript	Creates a responsive interface for entering crop and environmental parameters and displaying prediction results.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python, Flask	Handles web requests, input validation, model loading and crop prediction logic.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	CSV Dataset, Pickle Model	Stores crop training data and the serialized trained model used for prediction.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Render / Local Flask Server	Runs and hosts the Flask-based OptiCrop web application.
5	**Version Control & CI/CD**	Git, GitHub	Provides source control, collaboration and project repository management.

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	scikit-learn, pandas, NumPy, VS Code, Playwright	Supports machine-learning model development, data processing, coding and cross-browser application testing.